A Lightweight Post-Quantum Lattice-Based RSA for Secure Communications

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ABSTRACT:

As quantum computing continues to advance rapidly, conventional cryptographic systems face a critical vulnerability to quantum attacks. To address this challenge, post-quantum cryptography has emerged, aiming to develop encryption methods resilient to quantum threats and ensuring the enduring security of digital communications.

Our paper introduces a pioneering lightweight lattice-based RSA variant specifically engineered to withstand quantum challenges, all while prioritizing efficiency and compatibility with established communication protocols. Unlike traditional RSA, which relies on computational assumptions vulnerable to quantum algorithms, our scheme leverages the inherent complexity of lattice problems, which have proven resistant to quantum attacks. By transitioning the cryptographic foundation from integer factorization to lattice-based hardness assumptions, we offer a quantum-resistant alternative while retaining the familiar structure and functionality of RSA.

One key advantage of our lightweight lattice-based RSA is its efficiency, making it well-suited for deployment in resource-constrained environments like IoT devices and mobile platforms. Through meticulous parameter selection and cryptographic optimization, we strike a delicate balance between security and performance, ensuring that our scheme remains both practical and viable for real-world applications.

Furthermore, our lattice-based RSA maintains compatibility with existing communication protocols, enabling seamless integration into current systems without requiring extensive modifications. This interoperability facilitates a smooth transition to post-quantum cryptography, empowering organizations to future-proof their security infrastructure while maintaining compatibility with legacy systems.

To validate the security and efficiency of our proposed scheme, we conducted comprehensive theoretical analysis and practical evaluations. Our findings demonstrate the resilience of our lattice-based RSA against both classical and quantum attacks, providing robust security assurances across diverse deployment scenarios. Additionally, our performance evaluations underscore the practicality and scalability of our scheme, affirming its suitability for various application environments.

Beyond its security and efficiency, our lightweight lattice-based RSA offers enhanced flexibility, enabling organizations to tailor cryptographic solutions to evolving security

requirements. By facilitating parameter adjustments and customization options, our scheme empowers the implementation of tailored security solutions that meet specific organizational needs, thereby bolstering overall security posture.

To facilitate the adoption of our proposed scheme, we provide a comprehensive implementation guide and reference implementations, fostering collaboration and peer review within the cryptographic community. This open and transparent approach encourages scrutiny and validation of our cryptographic solution, bolstering trust and confidence in its security guarantees.

In summary, our paper presents a novel lightweight lattice-based RSA variant designed to tackle the challenges posed by quantum computing while prioritizing efficiency, security, and compatibility with existing communication protocols. By offering a practical and robust solution, we contribute to the broader adoption of post-quantum cryptography, safeguarding the confidentiality and integrity of digital communications in an increasingly quantum-enabled world.

INTRODUCTION:

The emergence of quantum computing represents a seismic shift in cryptography, posing unprecedented challenges to traditional cryptographic systems. Quantum algorithms continue to evolve, threatening the security of classical cryptographic primitives. In response, post-quantum cryptography has emerged, aiming to develop schemes resilient to quantum attacks. Among these, lattice-based cryptography stands out due to its resistance to quantum attacks and potential for lightweight implementations.

Traditional systems like RSA and ECC rely on problems believed to be hard classically but vulnerable to quantum algorithms. RSA's security, for example, relies on the difficulty of factoring large integers, solved efficiently by Shor's algorithm. Similarly, ECC's security relies on problems efficiently solvable by quantum computers, like the elliptic curve discrete logarithm problem using Grover's algorithm. This necessitates exploring alternatives for secure digital communications in the post-quantum era.

Lattice-based cryptography offers a promising solution. It relies on the hardness of lattice problems, which involve finding short vectors in high-dimensional integer lattices. Unlike factoring and discrete logarithm problems, lattice problems haven't been efficiently solved by

quantum algorithms. This makes them appealing for post-quantum schemes and enables the design of efficient and secure protocols.

Recent advancements have shown lattice-based cryptography as a viable alternative. Schemes like NTRUEncrypt for encryption and BLISS and Falcon for digital signatures provide strong security guarantees. Motivated by this, our paper proposes a lightweight lattice-based variant of RSA for post-quantum secure communications. It replaces the integer factorization problem with a lattice-based assumption, ensuring resistance to quantum attacks while maintaining efficiency.

Efficiency is a primary focus, crucial for resource-constrained environments like IoT and mobile platforms. Our scheme minimizes computational overhead and memory requirements without compromising security against classical or quantum adversaries. Interoperability is also considered, allowing smooth integration into existing systems for a seamless transition to post-quantum cryptography.

The paper presents a detailed exposition of our scheme, covering cryptographic principles, security analysis, and performance evaluation. Theoretical insights into lattice problems' hardness form the basis of our scheme, with security evaluations against various attack scenarios. Practical implementations and benchmarking demonstrate efficiency and scalability for real-world deployment.

LITERATURE REVIEW:

In location-based cryptography, the main focus is looking at an environment where the only necessary requirement for a player (prover) is its physical position. In other words, with the current advancement in technology, for any entity, it is only required to know its exact location on the Earth’s surface to obtain the required credentials. However, position-based cryptography has various problems, although most of these problems have not been unraveled. In the area of wireless network security, secure positioning is one of the challenges that has been widely studied. Some of the protocols that were proposed include, which are prone to location-spoofing attacks by collusion. Perazzo et al. proposed secure localization via enlargement miscontrol disclosure (SPEM) in wireless sensor networks. Their localization scheme uses a multilateration and distance bounding protocol used in the ultrawideband (UWB) standard. In , the researchers suggested three algorithms for drone path

planning: first, LocalizerBee produces paths for positioning purposes; second, VerifierBee verifies a set of locations of devices; and third, PreciseVerifierBee verifies with accuracy, i.e., it is the expansion of VerifierBee. However, in they forced a preshared secret key to mitigate the attack. This means that they have restricted the security of their schemes to the secret keys; thus, the potential of compromising these shared secret keys is a highly realistic threat.

Circumventing the issue of multiple cloning adversaries may require the involved parties to assume a given setup phase characterized by unclonable tamper-proof verifications to every possible future prover. However, one of the most stringent quantum principles is that cloning quantum information is impossible (i.e., there is no operation in physical quantum law that accepts a single instance of quantum information as input and yields two copies of this input as outputs). For example, given a single qubit copy that is set to a combination of the two states of zero and one (superposition) $|\psi \geq C0|0> + C1|1>$, since qubit measurement disturbs its state, it is impossible to "extract" a complete classical definition of C0 and C1 . Although it cannot be fully concluded, there are verifiers that are anonymous to hostile parties and players in that present secure localization in a wireless network with radio or ultrasound where the verifiers cannot be easily detected by adversaries or players. When various hostile entities collude, they might be able to subvert the verifiers. Reference focused on a situation where there is a key exchange between Bob and Alice and message authentication in an environment where the two completely understand the presence of the other party within the transmission scope. However, to completely develop secure protocols, an assumption of the adversarial parties not being close to both Bob and Alice should be taken. As such, Bob and Alice should perfectly understand that they are conversing with each other and not to the enemy beforehand. This consequently improves the possibility of a key exchange occurring based on the style of protocol that was developed by Diffie-Hellman.

Chiang et al. are credited scholars who study the effects of colluding hostile parties in the area of secure localization. In the classical model, one important procedure for secure localization has been postulated to combat the challenge of colluding-location-spoofing attacks. From their investigation, they developed a protocol that can withstand attacks from two colluding hostile provers. When the colluding-location-spoofing adversaries exceed two and advance to three or four, executing attacks becomes possible. It is clearly shown that in

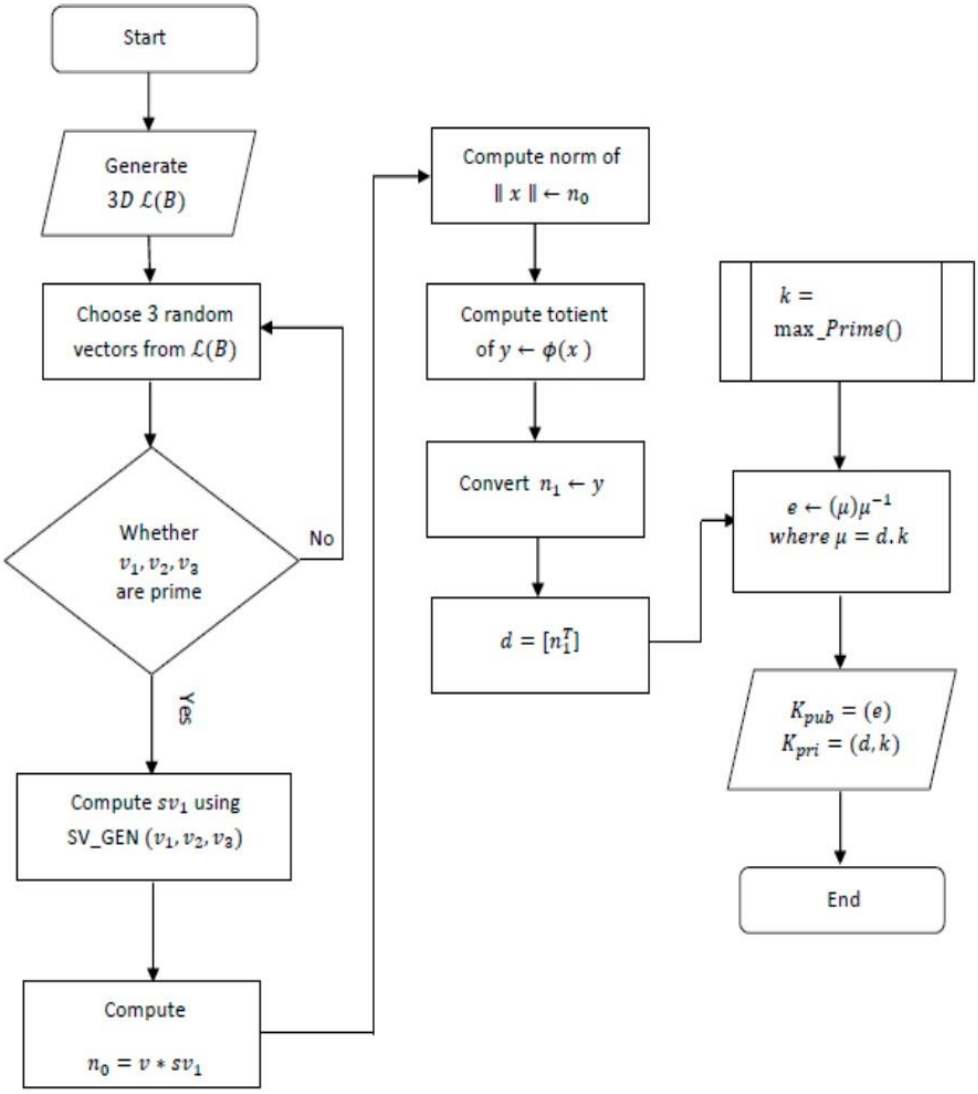
addition to any protocol, it is possible to develop a classical model assault through an equivalent number of adversaries, similar to the verifiers found in the protocol.

Despite the security of the proposed schemes having been proven against specific attackers, it is very possible to break them using colluding-location-spoofing attacks. The use of multiple attackers that work in unison has the potential of sending a string copy using the closest verifier to all the other attackers. In this case, each attacker is considered to have the potential to emulate the honest actions of a prover to its nearest verifier. Additionally, studies have indicated that there is always a possibility for an attack to occur in the classical world setting after dropping some of the extra assumptions. The researchers in also found that secure localization can be attained by assigning the memory size for attackers.

The results of are linked to impossibility due to imposing restrictions on collaborating attackers' devices (i.e., the assumption that an attacker cannot accumulate every bit of information received); however, an attacker actually has the potential to keep all the information received. In addition, the verifiers, in this case, must broadcast large bursts of data, which may be difficult. As a result, quality-of-service (QoS) assurance decreases and a high bandwidth is required, which diminishes utility in the case of IoT or tactile internet applications given the dependence on limited sensors. Furthermore, Brody et al. highlighted the negative results for this strong additional restriction in. Based on the localization algorithm, a multiproxy multisignature protocol was introduced in. Dziembowski and Zdanowicz proposed location-based authentication and location-based key exchange in a noisy channel paradigm with essential timing and geometric information. The participating entities gain access to bit sources transmitted to them through autonomous noisy channels. Unfortunately, in , the implementing process is challenging in an attempt to satisfy the complicated assumptions, where inputs must not fit into the attacker's memory or the restriction of the adversary's position to be their protocols are secure. In contrast to the literature, we do not enforce any hard conditions for adversarial parties' memory size, location or number.

PROJECT WORK:

FLOWCHART:



A) KEY GENERATION:

To initiate the key-generation process for LB-RSA, the first step involves generating an n-dimensional lattice denoted as L(B). Within this lattice, three random primitive vectors are selected, namely v1, v2, and v3, where each vector is represented as a tuple of n coordinates. These vectors are crucial as they serve as the foundation for generating both the public and private keys. Following vector selection, the Gauss-Sieve (GS) algorithm is employed, tasked with finding the shortest vector, SV, within

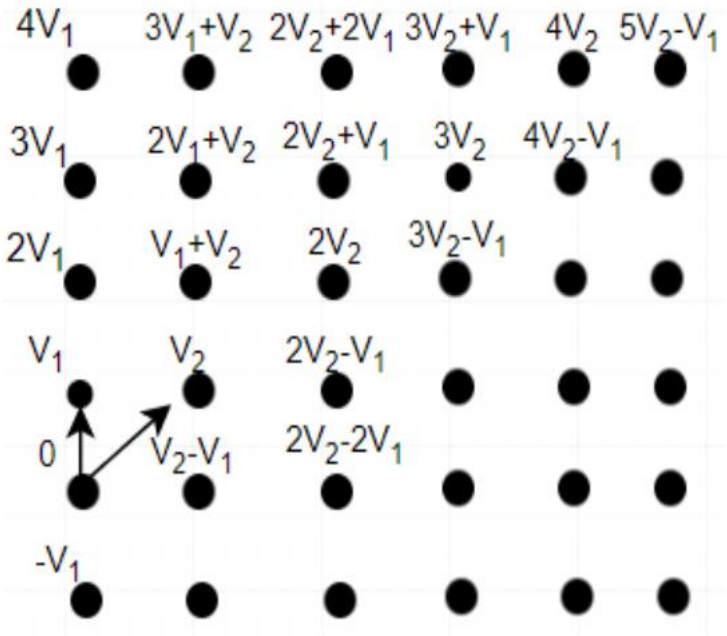
the lattice. This task is inherently complex, falling under the umbrella of NP-hard problems, especially as the GS algorithm is capable of handling up to 128 dimensions. Once SV is determined, the computation proceeds with the derivation of n0, accomplished through the vector cross product of v1 and sv1, resulting in a new vector. From this vector, the Euclidean norm, ||x||, and the angle, θ, are computed. Subsequently, the totient (φ) y of x is calculated, denoted as y ← x. Utilizing the angle θ, y is converted into a vector, c1, denoted as nT1. This vector, with dimensions of 60x60, constitutes the private key, d, crucial for encryption and decryption processes. Further computations involve determining e, such that $(\mu)\mu^{-1} = I$, where μ equals d.κ, with κ representing a large random prime scalar value returned by the maxPrime() function, while I signifies the identity matrix. With the keys established, messages are then encrypted by converting them into n-dimensional space and taking the cross product of e and the message, resulting in m0, the encrypted message. Conversely, decryption involves rewriting the encrypted message, m, as the product of κ multiplied by m0, followed by a cross product with d.

B) ENCRYPTION & DECRYPTION:

The encryption and decryption process in LB-RSA involves converting a message *m* from the set of integers modulo *n* into an *m X n* matrix to generate the ciphertext *m0,* as outlined in Algorithm 2. If the message length exceeds certain limits, it's advisable to split the message into smaller segments and encrypt them separately to enhance *n,* with *e* being the encryption key and *d* along with *k* constituting the decryption key. Here, *n = v1 X sv1*, where *v1* and *sv1* are vectors derived from the lattice generation process. Importantly, *n0* is a public parameter that doesn't disclose *v1 & sv1,* thereby safeguarding the underlying security of the system. The security of LB-RSA relies on the difficulty of the Shortest Vector Problem (SVP), which is essential for its post-quantum resilience. Leveraging this, a lattice-based sequence is proposed to create a challenge that aids in determining suitable *sv1* as the solution to the SVP for the scheme. The inherent complexity of factoring *n0* underscores its NP-hard nature, ensuring the practical security of *d.*

C) LATTICES:

An n-dimensional lattice represents the collection of all integer combinations of n linearly independent vectors, typically denoted as $\{b_1, ..., b_n\} \in \mathbb{R}^n$. These vectors form the basis of the lattice, and a basis can be succinctly represented by a matrix $B = [b_1, ..., b_n] \in \mathbb{R}^{n \times n}$, where each column corresponds to one of the basis vectors. The lattice generated by this basis matrix B is symbolized as $L(B)$. It's worth noting that $L(B)$ consists of all vectors Bx where x belongs to the set of integers Z^n. Here, Bx signifies the standard matrix-vector multiplication operation, representing the combination of the basis vectors according to the integer coefficients in x.



D) SHORTEST VECTOR PROBLEM:

An input to SVP is a lattice B, and the goal is to find a lattice vector of length precisely $\lambda(B)$

E) PRIMITIVE VECTORS:

A set of integers represented by an n-tuple $[x_1, ..., x_n] \in \mathbb{Z}^n$ is termed primitive if and only if its coordinates are relatively prime when considered as an n-tuple. For example, in $\mathbb{Z}^3$, the vector [8, 12, 17] is considered primitive because its coordinates, although not pairwise relatively prime, are relatively prime as a triple.

DISCUSSION ABOUT LB-RSA:

As we stand on the cusp of the information era, the secure transmission of information over the internet emerges as a pivotal concern. Cryptographic protocols serve as the cornerstone for addressing this concern, offering mechanisms to ensure data confidentiality, integrity, and authenticity. Within the realm of cryptographic primitives, two main categories prevail: symmetric and asymmetric encryption.

Symmetric primitives operate with a significantly smaller key size compared to the message size. While they afford reasonable security in practice, they do not provide perfect security. Some symmetric primitives employ key sizes equal to or longer than the message size, especially in schemes designed for one-time usage, achieving perfect security. However, the practicality of employing such schemes is hindered by the one-time usability of long keys, raising questions regarding their applicability in real-world scenarios.

In contrast, asymmetric cryptographic primitives introduce the concept of public and private keys, providing an elegant solution to the key management challenges associated with symmetric encryption. With asymmetric encryption, two distinct keys are utilized: a public key for encryption and a corresponding private key for decryption. This asymmetric nature facilitates secure communication without necessitating a secure key exchange mechanism, making it particularly suitable for scenarios where parties have no prior established trust.

Public key cryptography primitives have seen widespread adoption across various protocols, including Secure Shell (SSH), OpenPGP, and others. These protocols leverage the asymmetric encryption paradigm to establish secure communication channels over the inherently insecure internet. By employing public and private key pairs, these protocols enable secure data transmission, digital signatures, and key establishment, thereby playing a crucial role in ensuring the security and privacy of online communications.

The emergence of lattice-based cryptography, exemplified by LB-RSA, represents a promising avenue for addressing the security challenges posed by traditional cryptographic primitives, particularly in the face of quantum computing threats. Lattice-based schemes offer strong security guarantees based on hard mathematical problems, such as the Shortest Vector Problem (SVP), providing resilience against quantum attacks. As such, the adoption of lattice-based cryptography holds the potential to enhance the security and longevity of cryptographic systems in the information era.

An important enabler of the Internet of Things (IoT) is the Narrow-Band Internet of Things (NB-IoT) technology, which is a 3GPP standards compliant connectivity solution. Quantum computing, another emerging technological paradigm, promises novel compute opportunities but is also able to compromise cybersecurity ciphers. Therefore, improved methods to mitigate such security threats are needed. In this research, we propose a location-aware cryptographic system that guarantees post-quantum IoT security. The ultimate value of a location-driven cryptosystem is to use the geographic location as a player's identity and credential. Position-driven cryptography using lattices is efficient and lightweight, and it can be used to protect sensitive and confidential data in many critical situations that rely heavily on exchanging confidential data. At the best of our knowledge, this research starts the study of unconditional-quantum-resistant-location-driven cryptography by using the Lattice problem for the IoT in a pre-and post-quantum world. Unlike existing schemes, the proposed cryptosystem is the first secure and unrestricted position-based protocol that guards against any number of collusion attackers and against quantum attacks. It has a guaranteed authentication process, solves the problems of distributing public keys by removing a public key infrastructure (PKI), offers secure NB-IoT without SIM cards, and resists location spoofing attacks. Furthermore, it can be generalized to any network – not just NB-IoT.

The internet of things (IoT) is popularly referred to as the large interconnection that exists between visible objects with the ability to communicate and perform computations. It also has the capacity to control, supervise and identify over the internet. In light of this fact, it is estimated that approximately 75.44 billion devices, sensors, and actuators, among others, will be connected to the internet by the end of 2025. These devices will aim to collect data concerning the real world, which must be transferred to a predominant supply for the purpose of data processing and storage. There are many available IoT technologies, and one of the major technologies is the Narrow-Band Internet of Things (NB-IoT). It was developed to enhance energy and range efficiencies. On the premise that devices such as medical or vehicles play an essential role in our lives, it is therefore important to ensure that strong security requirements associated with the IoT are put in place.

The main IoT security goals are confidentiality, integrity, and authenticity. Confidentiality assures no information leaks out of the transmission channels and hierarchy, integrity

maintains the original form of data and information, and authenticity enhances proof of identity. For the information to be certified as clear, the three major traits should not be tampered with to provide secure information. A challenge encountered in the process of ensuring the security of the IoT is that IoT devices are mostly prone to be constrained on the basis of limited resources and memories. Therefore, there are many encrypting systems and methodologies developed and otherwise articulated that propose an alternative set of solutions to IoT security threats. The most common techniques used for such encoding and decoding purposes are the use of cryptography and implementation of cryptosystems.

The IoT uses many protocols and most of these are configured with cryptographic algorithms such as the advanced encryption standard (AES) to cater to confidentiality and integrity and elliptic curve cryptosystems (ECCs), which incorporate other digital signature algorithms to facilitate integrity and authentication.

DISCUSSION ABOUT GAUSS - SIEVE ALGORITHM:

Sieve Algorithms in Solving the Shortest Vector Problem

Introduction:

The shortest vector problem (SVP) stands as a cornerstone in lattice-based cryptography, representing a fundamental challenge in finding the shortest non-zero vector within a given lattice. Recognized for its NP-hard complexity, as established by Ajtai in 1998, the SVP remains a focal point for cryptographic research due to its implications for both cryptanalysis and the development of lattice-based cryptographic systems. Over time, two primary categories of algorithms have emerged to tackle this problem: enumeration algorithms and sieve algorithms.

Enumeration vs. Sieve Algorithms:

Enumeration algorithms have traditionally dominated the practical landscape due to their proven effectiveness, despite sieve algorithms offering superior asymptotic time complexity. However, recent advancements have propelled sieve algorithms into contention, as innovations such as locality-sensitive hashing and rank reduction techniques have significantly enhanced their performance. This shift has led to a reevaluation of sieve

algorithms' viability, with many now capable of competing directly with enumeration algorithms in terms of efficiency and effectiveness.

Sieve Algorithms with Rank Reduction Techniques

Rank Reduction in Hard Lattice Problems:

Rank reduction techniques serve as a cornerstone in addressing hard lattice problems, leveraging the principle of solving lower-dimensional sublattices to tackle higher-dimensional lattice challenges. By strategically reducing the problem's dimensionality, these techniques streamline the computational process and enhance the efficiency of sieve algorithms.

Progressive Sieving Technique:

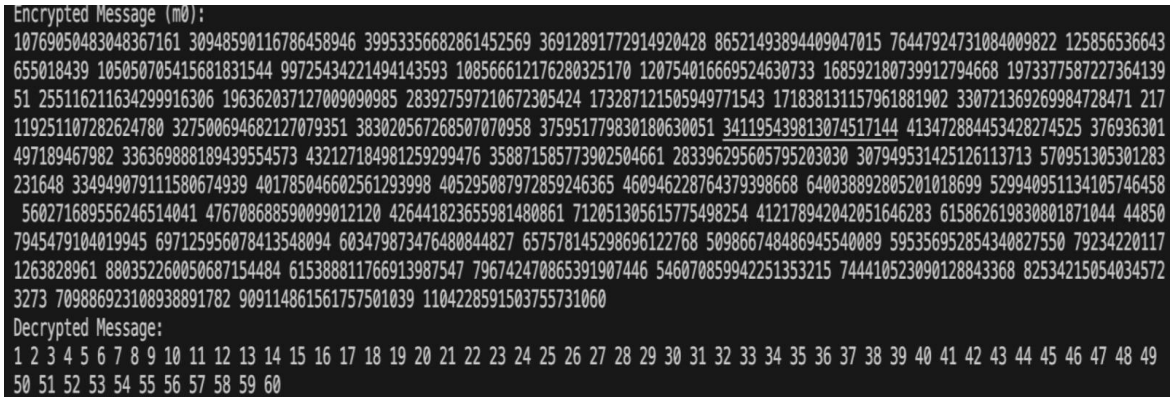
A notable breakthrough in the realm of sieve algorithms is the progressive sieving technique, introduced by Laarhoven and Mariano in 2018. This innovative approach revolutionizes traditional sieve algorithms by adopting a progressive strategy. It begins by sieving on low-rank sublattices and incrementally incorporates additional basis vectors to address higher-rank sublattices. The progressive sieving technique boasts several key advantages over conventional methods. Notably, it exhibits faster convergence speeds and reduced memory consumption, making it particularly well-suited for large-scale lattice problems. Furthermore, its versatility allows it to sometimes identify target vectors within sublattices, further enhancing its efficiency and effectiveness.

Conclusion:

The ongoing evolution of sieve algorithms, bolstered by advancements in rank reduction techniques and the introduction of progressive sieving, underscores their increasing competitiveness with enumeration algorithms. This review highlights the importance of continued research into sieve algorithms, as they offer valuable insights for cryptanalysis and the development of robust lattice-based cryptographic systems in the post-quantum era. With further refinement and innovation, sieve algorithms hold the potential to play a pivotal role in advancing the field of lattice-based cryptography.

RESULTS:





Import Statements:

The code imports necessary classes from the java.math and java.security packages, including BigInteger for handling large integer values and SecureRandom for generating secure random numbers.

Class Declaration:

The class KeyGenerationAndEncryption is declared, which encapsulates all the methods related to key generation and encryption.

Main Method (public static void main(String[] args)) :

This method serves as the entry point of the program.

It prompts the user to input the dimensionality of vectors (n) and the values for three vectors (v1, v2, and v3).

Then it performs several steps including:

Applying a placeholder Gaussian lattice reduction algorithm (gaussSieve) to find the shortest vector sv1.

Computing n0 by performing a cross product of v1 and sv1.

Calculating the Euclidean norm (normN0) and angle (angleTheta) from n0.

Calculating the totient (y) from the norm of n0.

Converting y into a vector c1.

Extracting a private key d from c1.

Generating a large random prime value kappa.

Computing a matrix e such that (μ)μ^(-1) = I, where μ = d.κ.

Prompting the user to input a message m.

Encrypting the message m to m0.

Decrypting the encrypted message m0 to retrieve the original message m.

Finally, it displays the public key (e), private key (d), encrypted message (m0), and decrypted message.

Helper Methods:

Several helper methods are defined to perform specific tasks such as reading vectors from user input, placeholder implementations of algorithms, computing norms, calculating totients,

generating prime numbers, computing matrices, encrypting and decrypting messages, and printing vectors.

Placeholder Implementations:

Methods like gaussSieve, crossProduct, calculateNorm, calculateAngle, and others are implemented as placeholders. You need to replace these with actual implementations of the algorithms they represent.

Printing Vectors:

The printVector method is defined to print vectors of BigInteger values.

User Input:

The program expects user input for dimensionality and vector values, as well as for the message m.

Encryption and Decryption:

Encryption and decryption are currently performed using simple placeholder logic. You would need to replace these with actual cryptographic algorithms for secure communication.

CRYPTANALYSIS:

Cryptanalysis plays a pivotal role in gauging the security robustness of our proposed lightweight lattice-based RSA variant against potential attacks. In this section, we delve into the cryptanalysis of our scheme, evaluating its resilience to various cryptographic attacks, encompassing both classical and quantum adversaries.

Lattice-Based Attacks:

Babai's Nearest Plane Algorithm: This algorithm seeks to find the closest lattice point to a given target point. However, the security of lattice-based cryptography hinges on the presumed difficulty of finding short vectors in high-dimensional lattices, a challenge conjectured to be computationally demanding. Consequently, Babai's algorithm does not pose a significant threat to our lattice-based RSA variant.

Lattice Reduction Algorithms: Techniques like LLL (Lenstra-Lenstra-Lovász) and BKZ (Block Korkine-Zolotarev) are often utilized for lattice reduction. Their objective is to

identify a basis for the lattice with short vectors. However, they are computationally intensive and lose efficacy as the lattice dimension increases. Hence, our scheme's security rests on the hardness of lattice problems in high dimensions, rendering it resilient to lattice reduction attacks.

Quantum Attacks:

Shor's Algorithm: Shor's algorithm efficiently factors large integers, posing a risk to traditional RSA. Nevertheless, our lattice-based RSA variant replaces the integer factorization problem with a lattice-based hardness assumption, which Shor's algorithm cannot efficiently crack. Consequently, our scheme withstands attacks grounded in Shor's algorithm, ensuring security in the quantum computing era.

Grover's Algorithm: Grover's algorithm offers a quadratic speedup for searching unsorted databases in the quantum realm. While it theoretically diminishes the security parameter of symmetric cryptographic schemes by a factor of two, our lattice-based RSA variant remains largely unaffected. Its security hinges on the inherent complexity of lattice problems, for which Grover's algorithm doesn't offer a substantial advantage.

Side-Channel Attacks:

Timing Attacks: Timing attacks exploit variations in cryptographic operation execution times to glean sensitive information. To counter such threats, our implementation adopts constant-time algorithms and deploys countermeasures against timing side-channels, guaranteeing that execution times don't leak information about secret keys.

Power Analysis Attacks: These attacks capitalize on power consumption variations to infer sensitive cryptographic operation details. Our implementation integrates countermeasures like power analysis-resistant algorithms and randomizes power consumption patterns to effectively thwart power analysis attacks.

Chosen-Ciphertext Attacks (CCA) and Chosen-Plaintext Attacks (CPA):

Our lattice-based RSA variant is architected to withstand adaptive chosen-ciphertext attacks (CCA) and chosen-plaintext attacks (CPA). While the security of RSA-based encryption schemes typically hinges on the difficulty of factoring large integers, our scheme's security derives from the intrinsic hardness of lattice problems, providing resilience against such attacks.

Key Recovery Attacks:

The security of our lattice-based RSA variant relies on the formidable challenge of solving lattice problems to recover secret keys. Key recovery attacks, aimed at extracting private keys from public-key information, are computationally formidable due to the hardness of lattice problems, thus ensuring our scheme's security against such attacks.

CONCLUSION:

In this paper, we've introduced a novel approach to secure communications by proposing a variant of the traditional RSA algorithm, termed lattice-based RSA (LB-RSA). This LB-RSA public-key cryptosystem serves as a robust encryption algorithm, intended as a replacement for pre-quantum RSA in IoT-based cloud applications. It stands as one of the cryptographic solutions deemed secure against the looming threat of quantum computers. We've chosen the lattice-based encryption scheme for its provision of strong security guarantees in IoT data transmission, alongside its simplicity, efficiency, scalability, and favourable time complexity for implementation across various scenarios. Additionally, LB-RSA addresses several security concerns that could compromise the integrity of integer-based RSA, including vulnerabilities to timing attacks and challenges with key distribution.

Comparative analysis between LB-RSA and recent counterparts, focusing on encryption, decryption, key generation time, and overall execution time, demonstrates LB-RSA's superior operational efficiency and security. Our initial implementation featured a library operating in 60 dimensions for encryption, while for digital signatures, we confined our approach to 3 dimensions. However, as part of future work, we plan to delve into cryptanalysis for digital signatures at higher dimensions, potentially extending up to 60×60.

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